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Submission date: 18-Nov-2025 10:03AM (UTC+0530)
Submission ID: 2819141154
File name: IEEE_Conference_Template__3.pdf (133.63K)
Word count: 2641
Character count: 16531

A Comparative Study of Multi-Heuristic V2V Computation Offloading Strategies for Autonomous Vehicle Networks

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***Abstract*—Vehicular Edge Computing (VEC) has become a critical enabler for autonomous vehicles executing latency-sensitive perception and control workloads. While Vehicle-to-Infrastructure (V2I) offloading remains the dominant paradigm, its reliance on dense roadside units and vulnerability to communication latency limit practicality in real urban settings. Vehicle-to-Vehicle (V2V) computation offloading provides a low-latency, infrastructure-free alternative where vehicles assist each other using locally available computational resources. However, determining appropriate offloading strategies is challenging due to mobility-induced link variability, heterogeneous processing capabilities, and strict deadline constraints. This paper focuses on a comprehensive comparative evaluation of five heuristic V2V offloading strategies, including greedy, speed-aware greedy, threshold-based, game-theoretic, and load-balancing under identical mobility, communication, and resource constraints. A controlled MATLAB-based simulation environment was developed using trajectories generated by VanetToolbox. Performance was analyzed across multiple dimensions, including task completion time, offloading ratio, robustness to mobility constraints, vehicle density, and task complexity. Results demonstrate that while greedy offloading yields maximal throughput, it aggressively consumes resources and is prone to mobility-induced failures. Game-theoretic and load-balancing strategies consistently provide superior fairness, energy efficiency, and stability, particularly under constrained service-time conditions. Threshold-based selection offers minimal decision latency but at a significant performance cost.**

***Index Terms*—Vehicular Edge Computing, Vehicle-to-Vehicle (V2V) computation offloading, Autonomous vehicle networks, Heuristic offloading strategies ,Greedy offloading ,Speed-aware greedy ,Threshold-based offloading ,Game-theoretic offloading, Load-balancing, Mobility constraints, Task completion ratio ,Energy efficiency, Urban vehicular networks**

I. Introduction

Autonomous vehicles handle several tasks that require heavy computation and quick response times. Examples include object detection, planning routes, and preventing collisions. Many vehicles, especially lower-cost models, do not have enough processing power to run all of these tasks locally. For this reason, offloading part of the computation to external resources has become an important approach to improve safety and responsiveness. Most existing work relies on Vehicle-to-Infrastructure (V2I) offloading, where vehicles send tasks to roadside units or edge servers. V2I can be helpful, but it comes with practical challenges. Deploying RSUs(Road Side Units) across a city is expensive, coverage is uneven, and communication delay increases as vehicles move farther from the infrastructure. These issues limit the usefulness of V2I for applications that require low delay, especially in busy urban areas. V2V computation offloading offers a different path. Modern vehicles often carry strong processors, which means nearby vehicles can share their computing resources when conditions allow. This can reduce latency and avoid dependence on fixed infrastructure. However, V2V offloading is not straightforward. Vehicle mobility changes rapidly, communication links are short-lived, and processing capability varies from one vehicle to another. The brief time during which two vehicles remain in range adds another layer of difficulty. Although several studies discuss vehicular edge computing, many focus on a single method and compare it only with simple baselines. This makes it hard to understand how commonly used heuristics differ when placed under the same conditions. A unified comparison is therefore needed.

A. Motivation

Urban intersections bring vehicles close to each other for short periods, creating conditions that are well-suited for short-range communication and cooperative processing. At the same time, the traffic flow at intersections is unpredictable, and

vehicles often interact only briefly. Studying how different offloading strategies work in such settings can help improve the design of practical V2V systems.

B. Research Objectives

This study aims to:

- Build a simulation environment using mobility traces from VanetMobiSim and the IDM-IM model.
- Implement five heuristic V2V offloading strategies:
  - Greedy
  - Speed-Aware Greedy
  - Threshold-Based
  - Game-Theoretic
  - Load-Balancing
- Compare their performance in terms of task completion, delay, energy use, sensitivity to mobility, effect of density, and effect of task size.
- Identify the trade-offs that shape the performance of each strategy.

C. Major Contributions

The contributions of this work are as follows:

- A unified comparison of five common V2V offloading heuristics under the same simulation conditions.
- Use of realistic mobility modeling and communication constraints, including channel capacity and service-time limits.
- A broad evaluation across mobility, workload size, processing load, and vehicle density.
- Practical insights into how each strategy behaves: greedy methods achieve higher completion but struggle under high mobility; game-theoretic and load-balancing approaches offer more stable and energy-efficient.
- performance; threshold-based methods remain simple but less effective.
- A baseline that can support future development of more advanced or hybrid V2V offloading techniques.

II. Related Work

Computation offloading in vehicular networks spans cloud and edge computing, distributed optimization, and V2V cooperation. Early work relied on Mobile Cloud Computing (MCC), where vehicles send tasks to remote data centers. MCC provides high processing power but suffers from high latency, congestion, and reliability issues, making unsuitable for time-sensitive autonomous driving tasks. Mobile Edge Computing (MEC) and Vehicular Edge Computing (VEC) place servers closer to vehicles to reduce delay. Still, MEC depends on dense infrastructure and stable connectivity, which is often unrealistic in urban areas, and performance drops when vehicles move out of RSU(Road Side Unit) coverage.

A. Distributed Optimization

Optimization methods aim to find the best offloading decisions. Game-theoretic approaches let vehicles choose strategies to maximize utility but require iterative computations that are too slow for real-time use. Convex, non-convex, and stochastic optimization minimize delay and energy but assume stable channels, which rarely exist in dynamic V2V networks. Reinforcement learning and MDP-based approaches can learn optimal policies online but need large datasets and long convergence times, limiting practical deployment.

B. V2V Computation Offloading

Recent studies explore offloading tasks to nearby vehicles. V2V offers low latency, infrastructure independence, and natural clustering at intersections. Some works use distance- or speed-aware metrics or probabilistic link stability to decide task allocation. However, most studies test only one algorithm against simple baselines, limiting understanding of how common heuristics compare under the same conditions.

C. Research Gaps

- There is no comprehensive comparison of multiple V2V heuristics, such as greedy, speed-aware, threshold-based, game-theoretic, and load-balancing strategies.
- Many studies oversimplify mobility constraints, ignoring service-time limits and the effects of speed, range, and trajectory changes.

This study addresses these gaps by evaluating multiple heuristics under realistic mobility, communication, and computation conditions.

III. Offloading Policies

We evaluate five heuristic strategies for selecting a service vehicle (SV) for V2V computation offloading. All policies assume that feasibility checks, i.e., communication delay, computation time, and service-time constraints, are already satisfied.

A. Greedy Offloading

The greedy strategy selects the SV that provides the minimum total delay:

$$j^* = \arg\min_j \left(T_{t,ij} + T_{c,ij}\right)$$

It prioritizes performance but often overloads the fastest vehicle, reducing fairness and stability.

B. Speed-Aware Greedy Offloading

To prevent offloading to fast-moving vehicles with short link durations, this policy adds a mobility condition Offload only if:

$$T_{ij} < T_{loc} \quad \text{and} \quad T_{s,ij} \geq T_{ij}$$

It balances delay reduction with mobility safety.

C. Threshold-Based Offloading

The task vehicle selects the first SV that satisfies a predefined delay or speed threshold:

$$T_{ij} \leq \theta \quad \text{or} \quad |v_i - v_j| \leq \delta$$

This method reduces decision overhead but may produce suboptimal selections.

D. Game-Theoretic Offloading

Vehicles maximize a utility function that trades off delay and energy:

$$U_j = w_1 \left(-T_{ij}\right) + w_2 \left(-E_{ij}\right)$$

The SV with the highest utility is selected. This approach improves fairness and energy efficiency but introduces minor computation cost.

E. Load-Balancing Offloading

This policy selects the vehicle with the shortest CPU queue or lowest assigned workload:

$$j^* = \arg\min_j Q_j$$

It prevents bottlenecks and offers stable performance under high density.

IV. Simulation Setup

This study evaluates the five offloading strategies in a controlled simulation environment that integrates realistic mobility modeling with MATLAB-based communication and computation analysis

A. Tools and Frameworks

MATLAB: Implements mobility parsing, delay computation, feasibility checks, and all offloading algorithms.

VANETToolbox: A Vehicular Network Simulator based on DES

B. Mobility Parameters

Vehicles follow intersection-based trajectories generated from the mobility traces with: Speed range: 20–60 km/h

Communication range: 300 m

Simulation duration: 100 s Service-time windows between vehicles are derived from relative displacement and velocity

C. Computational Settings

Each task vehicle generates a partially divisible task of size L. Service vehicles have heterogeneous processing power with CPU frequencies in the range 10–30 GHz. The task vehicle uses a lower local CPU frequency to reflect limited onboard resources.

D. Communication Parameters

Bandwidth: 10 MHz Transmit power: 20 dBm Noise density: -174 dBm/Hz Channel gain computed using distance-based path loss Communication delay is calculated as:

$$T_{t,ij} = \frac{D_{ij}}{C_{ij}}$$

where D_{ij} is the data size and C_{ij} is the link capacity.

E. Evaluation Metrics

The strategies are assessed using:

- Task completion ratio
- End-to-end delay
- Energy consumption
- Robustness under mobility
- Effect of vehicle density

F. Scenario Variations

Each task vehicle generates a partially divisible task of size L. Service vehicles have heterogeneous processing power with CPU frequencies in the range 10–30 GHz. The task vehicle uses a lower local CPU frequency to reflect limited onboard resources. To examine behavior under different conditions, the following parameters are varied independently:

- Number of vehicles: 4 to 20
- Task size: small, medium, large
- Mobility level: low, moderate, high
- CPU load/queue length

V. Results and Key Findings

The following results were obtained from this study in terms of Overall Task Completion, ffect of Vehicle Density, Impact of Task Size, Mobility Constraints and Energy Consumption

A. Overall Task Completion

Greedy offloading achieves the highest task-completion ratio because it always picks the vehicle with the strongest processing capability. However, this aggressive selection pattern often overloads these high-performance vehicles, which reduces stability over longer periods. The game-theoretic and load-balancing strategies provide more stable completion rates, particularly when mobility increases. Threshold-based selection performs the weakest overall because it accepts service vehicles too early, often before better options become available.

B. Effect of Vehicle Density

Adding more service vehicles improves performance for all strategies at first, as the offloading opportunities naturally increase. However, the benefits level off once the number of vehicles reaches around 15. Beyond this point, higher communication contention and increased movement variability reduce overall gains.

- Greedy continues to select the fastest nodes, which can lead to occasional queue buildup.
- Load-balancing performs the best at high density because it spreads the workload more evenly.
- Game-theoretic remains stable but tends to be slightly conservative when choosing service vehicles.

C. Impact of Task Size

For small tasks, local processing is usually more efficient, as communication delays outweigh the benefits of offloading. In contrast, large tasks benefit significantly from offloading. Greedy and game-theoretic approaches show the most improvement for large workloads because they consistently choose vehicles with strong processing power. Threshold-based offloading struggles with larger tasks since its fixed selection rule does not adjust to changes in task complexity.

D. Mobility Constraints

Mobility has a major influence on V2V offloading performance. When the effective service-time window drops below roughly 1.3 seconds, all strategies experience reduced success due to unstable links. Speed-aware greedy handles mobility the best because it filters out vehicles moving too quickly

E. Energy Consumption

Game-theoretic offloading achieves up to 20% lower energy usage by avoiding selections that require long transmission durations. Load-balancing also reduces energy consumption by spreading tasks across several vehicles and preventing bottlenecks. Greedy offloading consumes the most energy because it repeatedly selects high-performance, and sometimes distant, vehicles that demand more communication effort.

VI. Discussion

The comparison of the five strategies provides several practical observations for designing usable V2V computation offloading systems. The first finding is that no single method works best in every situation. Greedy offloading reduces delay but performs poorly when vehicles move quickly and tends to use more energy. Similarly, the game-theoretic and load-balancing approaches offer more stable and fair performance, making them better suited for real environments where both resource distribution and link reliability matter. A second key insight is that mobility remains the main factor limiting V2V offloading. Even vehicles with high processing power cannot contribute effectively if changing speeds reduces the available service time. Strategies that account for relative speed or link duration, such as speed-aware greedy and game-theoretic selection, usually achieve higher overall success because they better match tasks to vehicles that can remain in range long enough. Third, the study shows that task size and communication overhead strongly affect how useful offloading can be. Small tasks are usually completed faster when handled locally. Larger tasks, however, gain from offloading as long as the communication link stays stable and does not add excessive delay. Finally, increasing vehicle density creates more offloading opportunities at first, but the benefit does not continue indefinitely. Beyond a certain point, link contention and unpredictable movements reduce overall performance. In these conditions, strategies that spread tasks across multiple vehicles, such as load-balancing, become more effective. Overall, the results show that successful V2V offloading requires balancing delay, mobility, energy use, and queue behavior together. Optimizing for only one factor is not enough to sustain performance in real traffic environments

VII. Conclusion

This study presented a comparative analysis of five heuristic strategies for V2V computation offloading in autonomous vehicular networks. Using realistic traces and a unified simulation framework, we evaluated greedy, speed-aware greedy, threshold-based, game-theoretic, and load-balancing strategies under different mobility, density, and workload conditions. The findings conclude that greedy offloading delivers the lowest delay but is highly sensitive to mobility and energy consumption. Game-theoretic and load-balancing strategies offer more stable performance, better energy efficiency, and greater reliability to dynamic link conditions. Offloading doesn't help much with small tasks. Bigger tasks, however, perform better when the communication overhead stays low.. Vehicle density improves performance up to a moderate threshold before diminishing returns appear. Overall, V2V offloading can support V2I and cloud systems effectively, especially in busy city intersections. But to keep performance stable, the methods must handle mobility changes, balance resources fairly, and manage energy carefully.

References

[1] C. Chen *et al.*, "Delay-Optimized V2V-Based Computation Offloading in Urban Vehicular Edge Computing Networks," *IEEE Access*, vol. 8, pp. 18863–18873, 2020.

[2] H. Hartenstein and K. Laberteaux, *VANET: Vehicular Applications and Inter-Networking Technologies*. Wiley, 2010.

[3] M. Chen *et al.*, "Vehicular cloud computing: A survey," *IEEE Trans. on Intelligent Transportation Systems*, vol. 16, no. 3, pp. 1243–1258, 2015.

[4] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing," *IEEE Commun. Surveys & Tutorials*, vol. 19, no. 4, pp. 2322–2358, 2017.

[5] X. Chen *et al.*, "Efficient multi-user computation offloading for mobile-edge cloud computing," *IEEE/ACM Trans. on Networking*, vol. 24, no. 5, pp. 2795–2808, 2016.

[6] K. Zhang *et al.*, "Energy-efficient offloading for mobile edge computing in 5G heterogeneous networks," *IEEE Access*, vol. 4, pp. 5896–5907, 2016.

[7] J. Liu, H. Guo, and H. Ding, "Load balancing in vehicular edge computing," *IEEE Trans. Vehicular Technology*, vol. 69, no. 3, pp. 2886–2898, 2020.

[8] Saitam, A. Mellouk, and S. Zeadally, "VANET-cloud: A cooperative architecture for mobile cloud computing in VANETs," *IEEE Trans. Vehicular Technology*, vol. 64, no. 12, pp. 5447–5459, 2015.

[9] S. Shladover, "Cooperative vehicle-highway automation systems," *IEEE Trans. Intelligent Transportation Systems*, vol. 12, no. 4, pp. 126–136, 2011.

[10] K. Abboud, H. A. Omar, W. Zhuang, "Interworking of DSC-based V2V and cellular LTE-V networks," *IEEE Trans. Vehicular Technology*, vol. 65, no. 12, pp. 9457–9470, 2016.

[11] C. Campolo, A. Molinaro, and A. O. Berthet, "Modeling V2V communications for cooperative awareness," *IEEE Commun. Magazine*, vol. 55, no. 8, pp. 80–86, 2017.

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